

Hydrogenic Orbitals

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1 Abstract

A hydrogenic atom is an atom or ion with one electron. Examples of hydrogenic atoms are H, He^+ , Li^{2+} [1]. We can calculate the probability density at a specific point in space by solving the Schrödinger equation and plotting the points in space where the probability density is the same will reveal the shape of an orbital.

Hydrogenic atoms are important because, as there is no repulsion between multiple electrons, the Schrödinger equation can be solved analytically [2]. Therefore, hydrogenic orbitals help us understand the shapes of orbitals in any atom, which is how the periodic table is arranged (the highest energy subshell of an atom decides which block the atom is placed in the periodic table) [3].

2.1 The History of the Atom

In 1913, Niels Bohr suggested that electrons orbited the nucleus and were held in electron shells. Electrons cannot occupy space between the electron shells. However, the development of wave-particle duality raised questions about Bohr's model [3].

In 1924, Louis de Broglie proposed that all objects in motion act as both particles and waves. The wave nature of electrons suggests that a single electron is “spread out”, which contradicts Bohr's model that states that electrons orbit in precise circular paths [5].

Additionally, Heisenberg's Uncertainty Principle states that it is impossible to measure both the position and the momentum of a subatomic particle simultaneously. Although knowing the exact values is impossible, approximations can be made, which are expressed as probabilities [5].

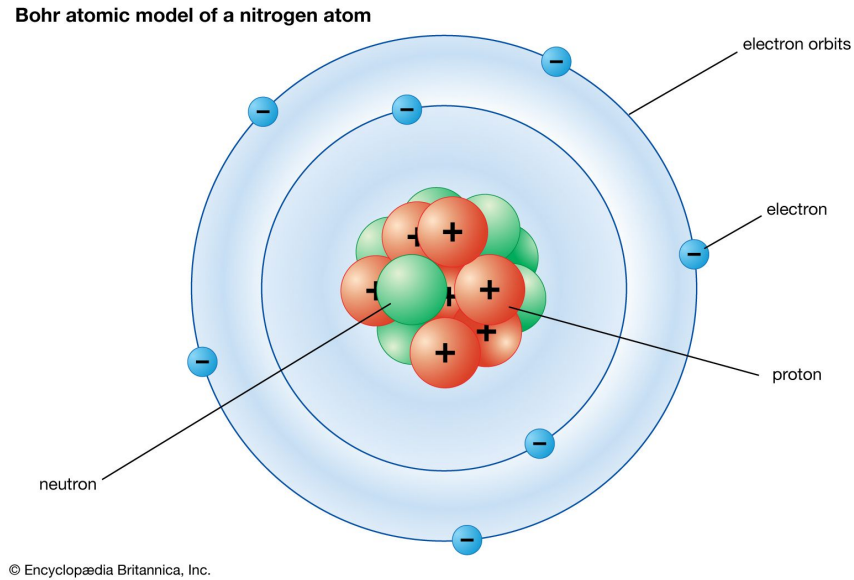


Figure 1: The Bohr model of an atom [4].

2.2 The Schrodinger Equation and Probability Density

An electron in an atom can be thought of as a circular standing wave surrounding the nucleus. This explains why electrons can only be found at specific energy levels, as a circular standing wave must have an integer number of wavelengths in order to exist. As an increasing number of wavelengths means the wave possesses more energy, higher electron shells have a higher energy [6].

In 1926, Erwin Schrödinger derived a formula which describes the system's wave function in quantum mechanics. According to the Schrödinger equation, electrons exist as a waveform, forming an electron cloud. The electron could be detected anywhere in the cloud but is more likely to exist in a specific region. [6]. The peak probability of finding an electron is the same distance as predicted by the Bohr model [5].

The time-independent Schrödinger equation is:

$$-\frac{\hbar^2}{2m}\nabla^2\psi + V\psi = E\psi$$

$\hbar$ = reduced Planck's constant, which is Planck's constant divided by 2

m = mass

$V(x)$ = potential energy calculated using Coulomb's law and using Z , the atomic

number

E = total energy

Ψ = wavefunction, which represents the quantum state of the particle based on its position [7]

$|\Psi|^2$, the absolute square of the wavefunction, tells us the probability that an electron can be found in a certain spot (because intensity is proportional to the amplitude squared) [6].

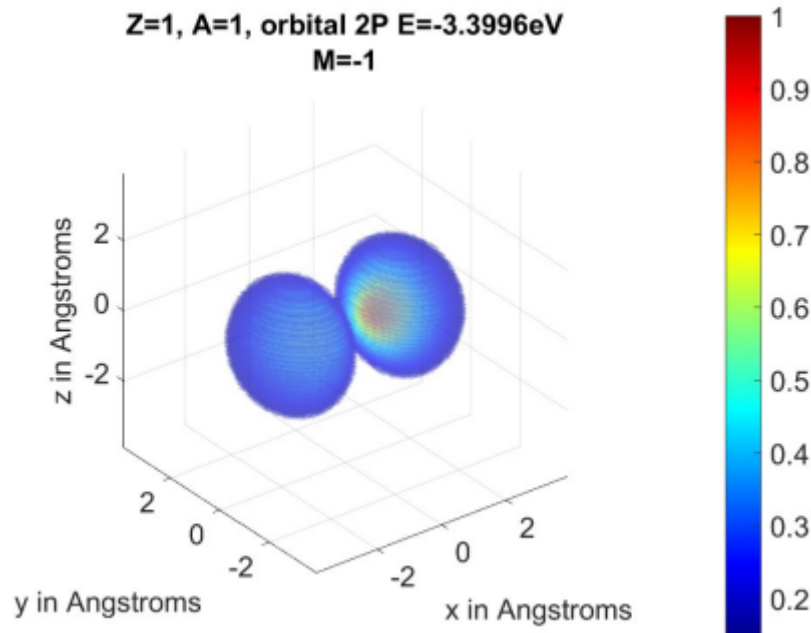


Figure 2: The probability density for a specific orbital plotted in 3d [7].

The wavefunction uses polar coordinates rather than cartesian coordinates. It is split into the radial wavefunction (which calculates how the electron's probability changes as the distance from the nucleus changes) and angular wavefunction which calculates how the electron's probability changes when the angle from the nucleus changes. The Radial Wavefunction is influenced by Z , the atomic number which states the charge in the nucleus, while the angular wavefunction is identical for all hydrogenic systems [7].

2.3 Quantum Numbers

The quantum numbers emerge from the Schrödinger equation. There are four quantum numbers: n , l , m_l and m_s [8].

The principal quantum number “ n ” represents the energy level of an electron. It can be any positive integer [8].

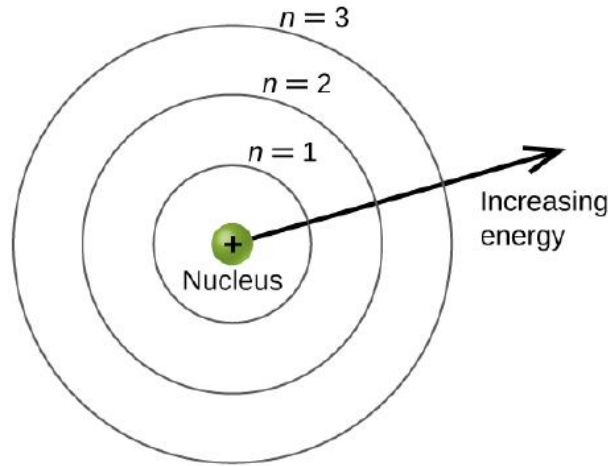


Figure 3: Electron shells described by quantum number n [9].

The angular momentum quantum number “ l ” describes the shape of the orbital. It can take any value from 0 to $n-1$ [8]. $l=0$ corresponds to an s-orbital, which has a spherical shape; $l=1$ corresponds to a p-orbital, which has a dumbbell shape; $l=2$ corresponds to a d-orbital and $l=3$ corresponds to an f-orbital [3].

The magnetic quantum number “ m_l ”, which describes the specific orbital from the subshell. It can take any value from $-l$ to l [8]. This means that there is 1 s-orbital, 3 p-orbitals, 5 d-orbitals and 7 f-orbitals in each shell [3].

The spin quantum number “ m_s ”, which describes the direction of spin. It can take the values of $+1/2$ or $-1/2$. For multi-electron systems (not hydrogenic) if there are two electrons in an orbital, then they have opposite spins. This allows the electrons to fulfil the Pauli Exclusion Principle [8]. The Pauli Exclusion Principle states that two or more fermions (an electron is a fermion) cannot simultaneously occupy the same quantum state [11].

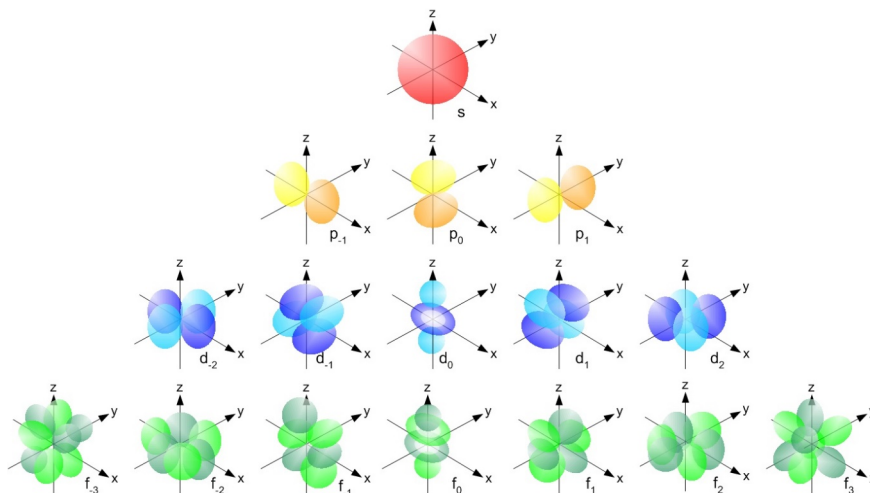


Figure 4: The Orbital shapes, based on the quantum numbers l and m_l [10].

3 Conclusion

Despite hydrogenic orbitals being an idealization, they allow us to understand multi-electron atoms, as unlike multi-electron atoms, they can be solved analytically instead of using approximations [2].

The quantum model of an atom also has real world applications, such as transistors and microprocessors, quantum computers, and molecular modelling [12].

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